

Paris at Ten

The world's carbon intensity is falling faster than ever. We ask whether the record can credit the Paris Agreement — and find that it cannot, for a specific and instructive reason.

True world aggregate, 1965-2023 · 44 specifications · unknown-break-date tests

THE QUESTION

Something changed. Was it Paris?

The good news is real. Global CO₂ growth fell from +2.1% a year over the Rio-to-Paris quarter century to +0.7% a year since, and carbon intensity is now improving at -2.1% a year: the fastest sustained rate in the observed record.

But the obvious reading — that the agreement did it — is the reading the data is worst at supporting. Carbon intensity has been falling for fifty years, in episodes with their own causes. Clean-energy costs collapsed on almost the same schedule as the treaty. And emissions growth slows when the economy slows.

So we ask a narrow question: when did the trend actually turn, and is 2015 a special year in the record — or a year that happens to sit near one?

Why we look at intensity

Emissions have three moving parts:

economic growth + fuel mix + energy efficiency

Growth pushes emissions up; the fuel mix and efficiency are the only two things that push them down. A policy effect has to show up in one of them before it can show up in emissions at all.

Four steps

1

Use a true world series

Not a sum over whichever countries reported. The published world totals for CO₂ and energy, 1965-2023, with World Bank world GDP. No composition drift, no interpolation, eight post-Paris years.

2

Let the data pick the turn

Fit two straight lines meeting at a kink, try every possible kink year, keep the best. Then a formal unknown-break-date test with bootstrapped critical values, so searching over years does not manufacture significance.

3

Ask if the treaty year is special

A curving trend makes almost any year look significant. We rank 2015 against every other candidate year — a treaty earns credit only if its year beats its neighbours.

4

Re-run everything 44 ways

World aggregate or country panel, four start years, three endpoints, four completeness thresholds, with and without interpolation. We report only what survives.

A turning point tells us when a trend changed — never, by itself, why. That is the whole method.

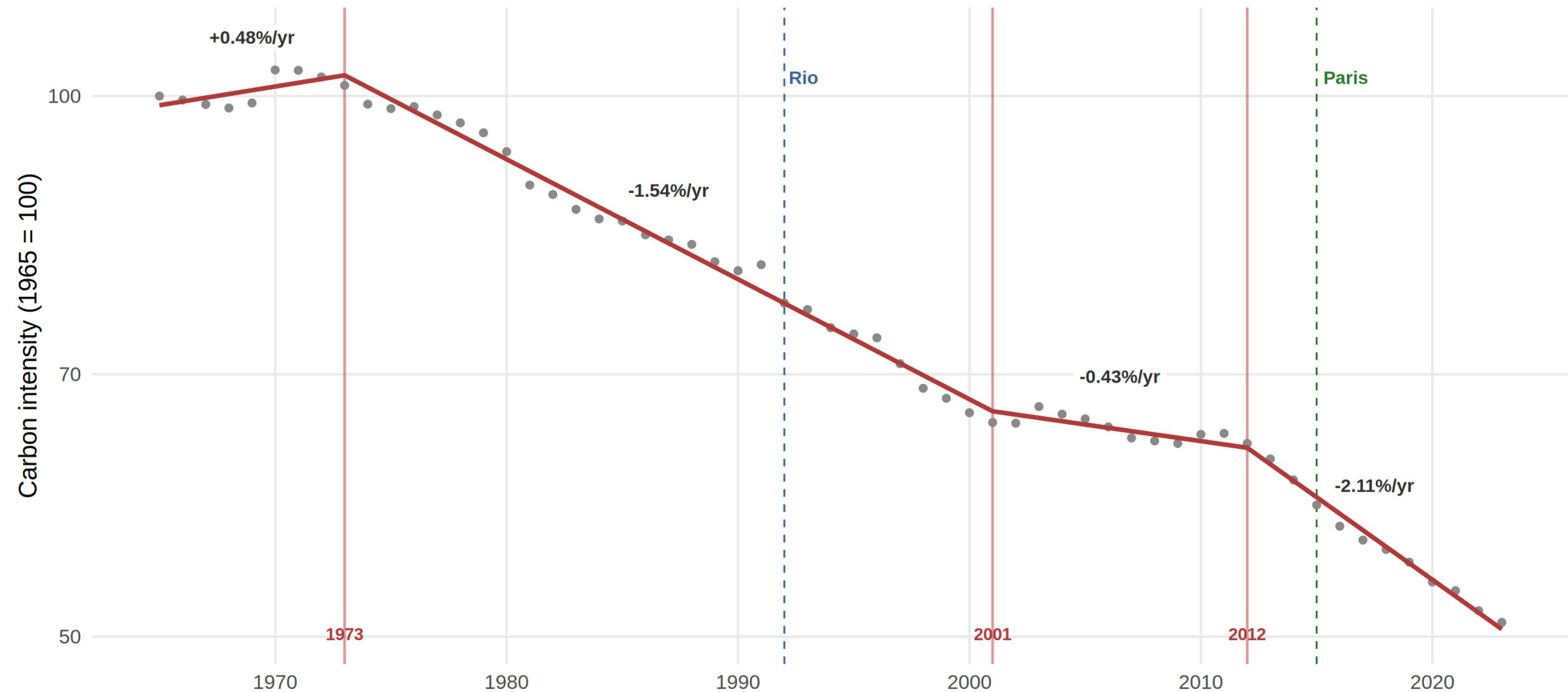
The decisive turn in the record is the 1970s — not a treaty

Takeaway: The oil shocks remain the largest structural break in the world's carbon intensity, and the current fast episode began in 2012.

The world's decisive turn was the 1970s oil shocks - and it is still the largest in the record

Carbon intensity of the true world aggregate (OWID world CO2 and energy; World Bank world GDP), 1965 = 100, log scale.

Red line: best multiple-break fit, breaks chosen by BIC. A formal unknown-break test puts the single dominant break at 1972 ($p = 0.002$).



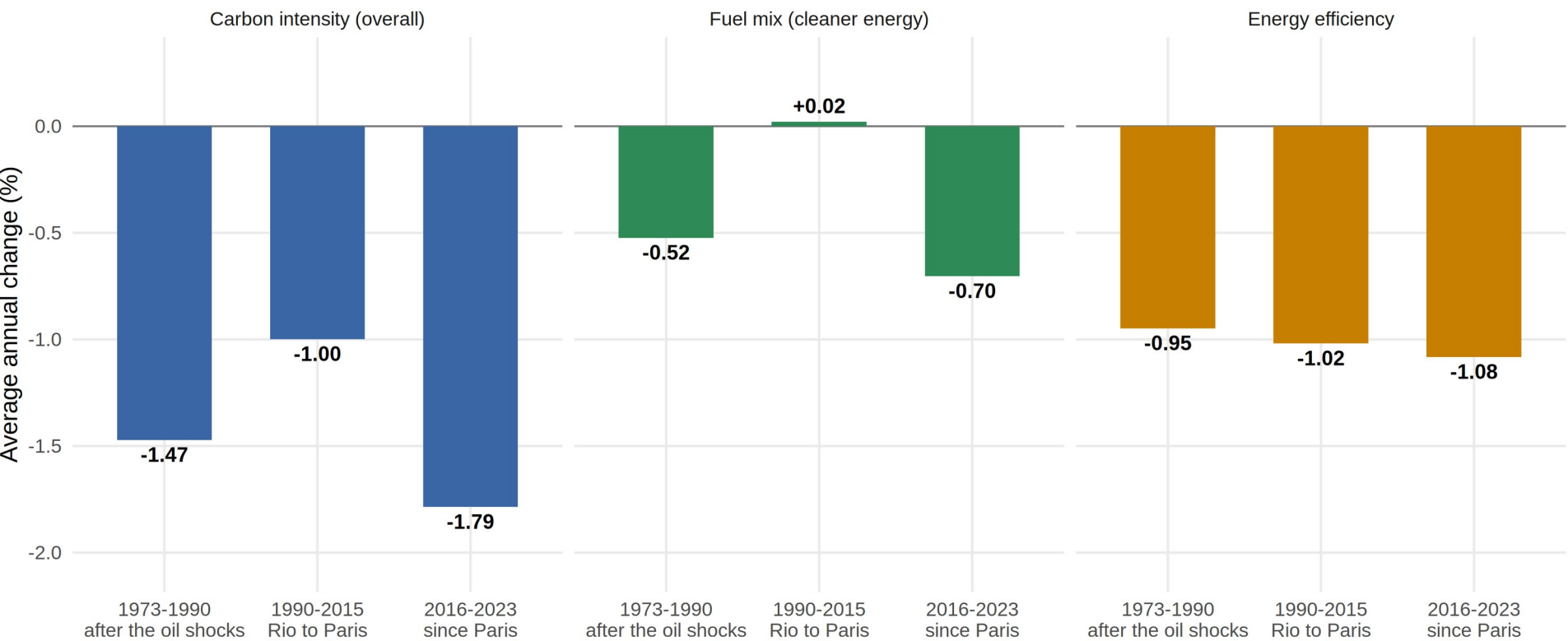
Efficiency never changed pace. The fuel mix stalled, then resumed.

Takeaway: Energy efficiency has improved at about 1% a year for fifty years. The whole story is the fuel mix, which stalled completely from Rio to Paris.

But: this is why the size of the 'Paris effect' depends on the baseline: against the 2000s stall it looks dramatic, against the post-1973 record it is a resumption.

Efficiency never changed pace. The fuel mix stalled for 25 years, then resumed.

Average annual change (%) in each component of the true world aggregate. Overall carbon intensity is now improving at its fastest rate on record - but that is the fuel mix resuming after a generation-long stall, not efficiency speeding up. How large the recent gain looks depends on which era you compare it to.

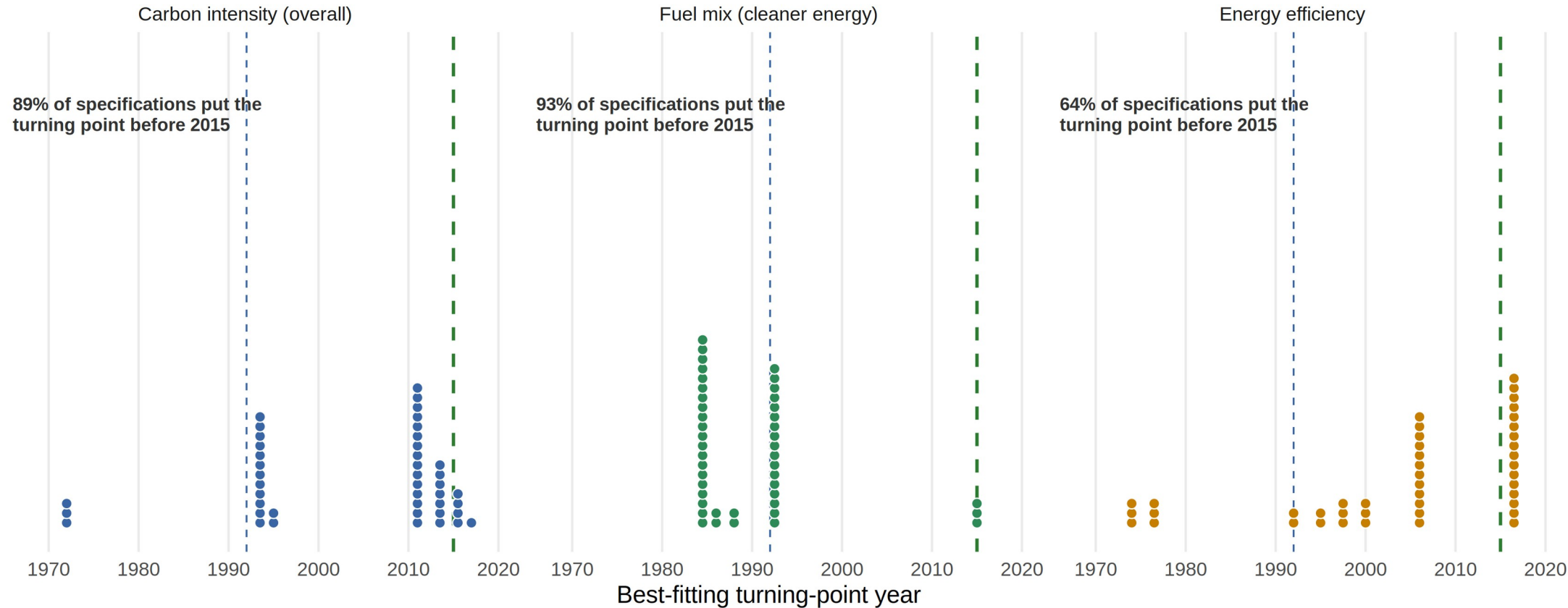


However we build the series, the turn comes before Paris

Takeaway: In roughly nine specifications out of ten, the best-fitting turning point lands before 2015 — usually 2011 to 2013.

Across 44 ways of building the series, the turning point lands before Paris

Each dot is one specification: true world aggregate or constructed country panel, four start years, three endpoints, four completeness thresholds, with and without interpolation. The dot shows the best-fitting turning-point year under that specification.



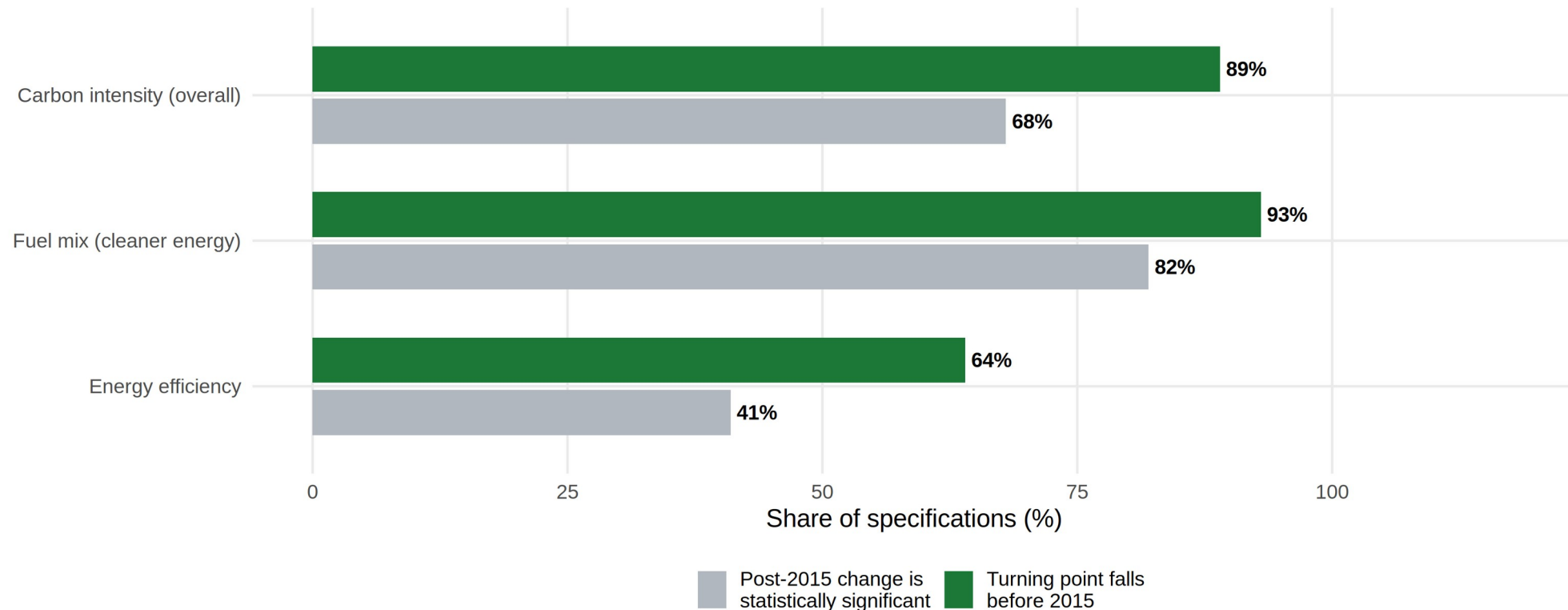
The timing result is robust. The rest is not.

Takeaway: Whether the post-2015 change is statistically significant depends on the component and on how the series is built — for efficiency, only 41% of specifications agree.

But: and Paris does not cleanly beat its neighbouring years: on the true world aggregate it ranks 80th percentile for overall intensity but only 11th for the fuel mix.

One finding survives every specification. The others do not.

Share of the 44 specifications in which each statement holds. The timing result is near-universal; whether the post-2015 change is statistically significant depends on the component and on how the series is built - especially for energy efficiency.



WHAT IT ADDS UP TO

A real acceleration that began before the treaty

What we can say

The world is decarbonizing faster than at any point in the record: carbon intensity has fallen at 2.1% a year since 2012, against 0.4% in the decade before. That improvement is real, it is large, and it is concentrated in the fuel mix — exactly where clean technology and energy policy operate.

What we cannot

The turn is dated to 2012 in almost every specification — three years before the agreement. Energy efficiency did not respond at all. The fuel-mix improvement is a return to its pre-1990 pace rather than something new. And the timing coincides exactly with the collapse in clean-energy costs.

The acceleration is real. It started before Paris, and it looks like what cheap clean energy would produce.

WHAT WOULD SETTLE IT

Three things — and two are not at the world level

Persistence. A 2012 turn driven by a one-off technology shock should decay as the cheap-solar transition matures. One that reflects policy ratcheting should not. Another decade of data distinguishes them; the size of the change is already detectable, so this is about durability rather than power.

The demand side. Efficiency is the informative half precisely because it has not moved. Cheap solar explains a cleaner fuel mix; it does not explain how much energy the world uses per unit of output. If efficiency accelerates, technology alone stops being a sufficient explanation.

Disaggregation. Paris works through national pledges, so its signature should appear country by country — where series are independent, placebo tests are genuinely separate, and a documentary record exists for what accompanied each turn. That is the companion paper.

The question we would put to the field: should treaty assessment be built on intensity turning points and placebo discipline, rather than on emissions levels and anniversaries?

LIMITS

What we are not claiming

This is not causal

A turning point identifies when a trend changed, never why. Even a perfect fit at 2015 would be a coincidence of timing. And an agreement that prevented backsliding would leave no turning point at all — absence of one is not absence of an effect.

Eight post-Paris years

The series ends in 2023 for CO2 and energy and 2022 for GDP. Country-level work stops in 2021 because GDP reporting, not emissions reporting, runs out. Adding partial years would overstate the recent trend.

Two GDP concepts

Overall intensity and efficiency use World Bank world GDP in constant 2015 dollars; the country panels use purchasing-power GDP. The fuel-mix result — the one most relevant to policy — uses a single source and no GDP at all.

Production-based accounting

Emissions are counted where they are produced. At the world level that is the right frame, since global production equals global consumption — but it matters for the national work that follows.

The curve is bending faster than ever.

It started bending in 2012.

- The largest break in the world's carbon-intensity record is still the 1970s oil shocks.
- The current fast episode begins in 2012 — in nine specifications out of ten, before Paris.
- Energy efficiency has improved at about 1% a year for fifty years and did not respond to the treaty era.
- The fuel mix stalled from Rio to Paris and has since resumed — which is what cheap clean energy would produce.

Turning points say when · the Kaya split says through which door · placebo tests say whether the date is special · the specification grid says whether any of it holds